

Dart on Arm

Flutter Bangalore Meetup - June 2021



A brief introduction



Engineer, The @ Company,
building a platform that puts people in
control of their data.

Co-host, Tech Debt Burndown Podcast

Cloud Editor, InfoQ

Links to socials etc. at chris.swanz.net



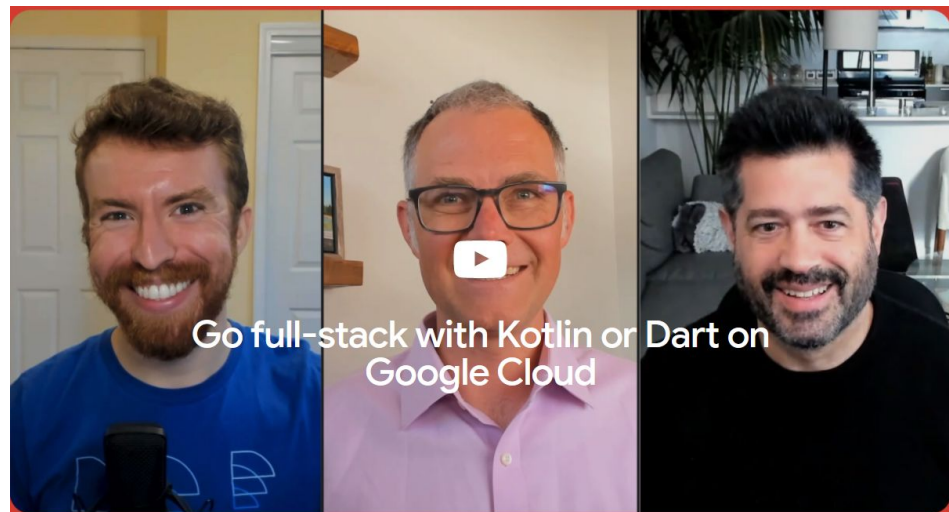
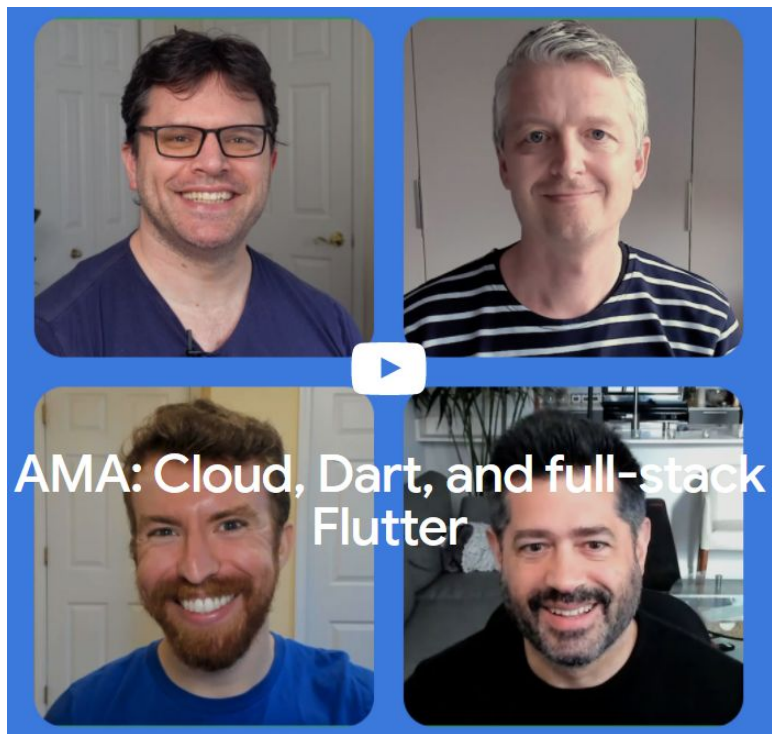
Agenda

- Start with WHY - why Arm is important
 - ◆ 'Full Stack Dart' to complement Flutter Apps
 - ◆ Arm growth in the cloud and IoT
- Just In Time (JIT) vs Ahead of Time (AOT) - the trade offs
- Dependencies for building and running AOT binaries
- All a bit x86 - how do I get Arm?
- Cross platform building made easy
- Q & A



Full Stack Dart

<https://events.google.com/io/session/083b441e-4076-4443-9750-6529b34257b7?lng=en>



<https://events.google.com/io/session/6d5795f2-ec1c-4c7a-9efa-ea90c8c59c03?lng=en>



Full Stack Dart

with Tony Pujals and Kevin Moore

<https://gcppodcast.com/post/episode-261-full-stack-dart-with-tony-pujals-and-kevin-moore/>

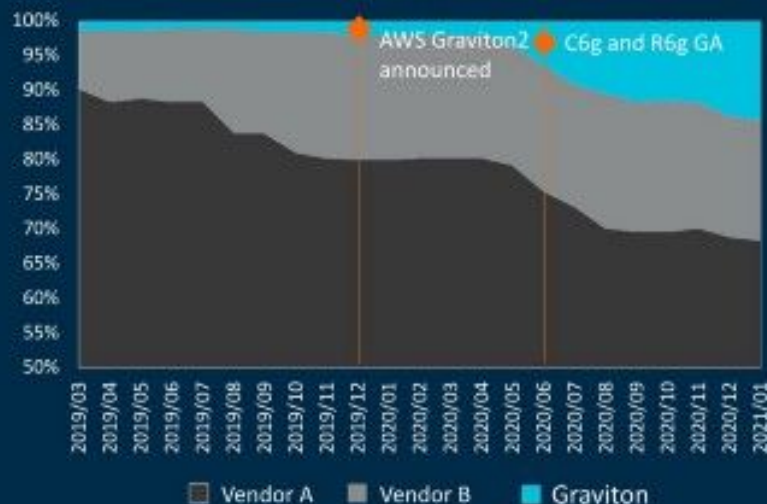
Arm growth is huge
in the cloud and IoT



AWS Graviton2 is Quickly Expanding its EC2 Footprint

Steady growth and regional expansion since mid-2020

AWS Instance Type Share

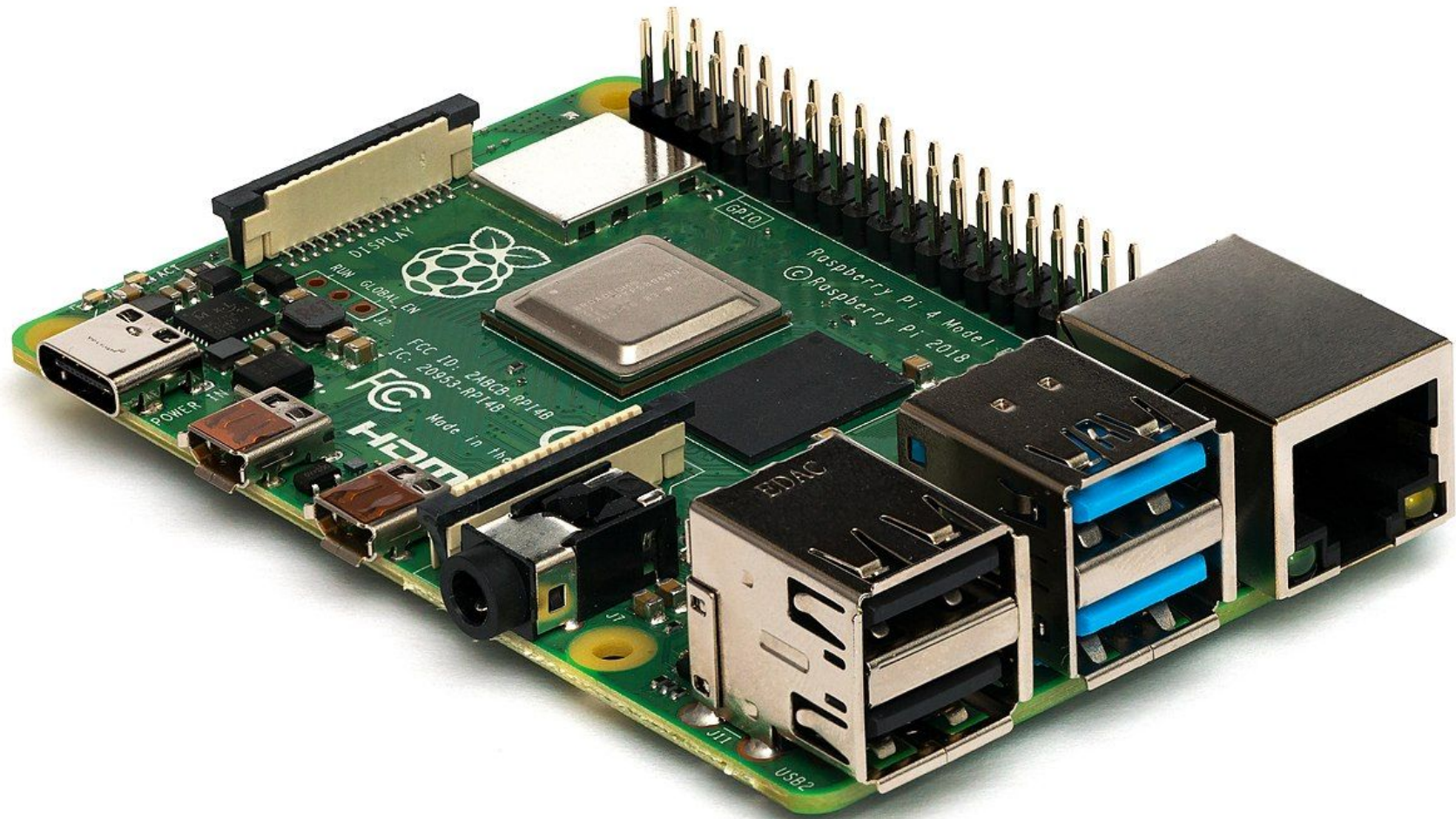


AWS Graviton2-based EC2 instances make up
14% of the installed base within AWS

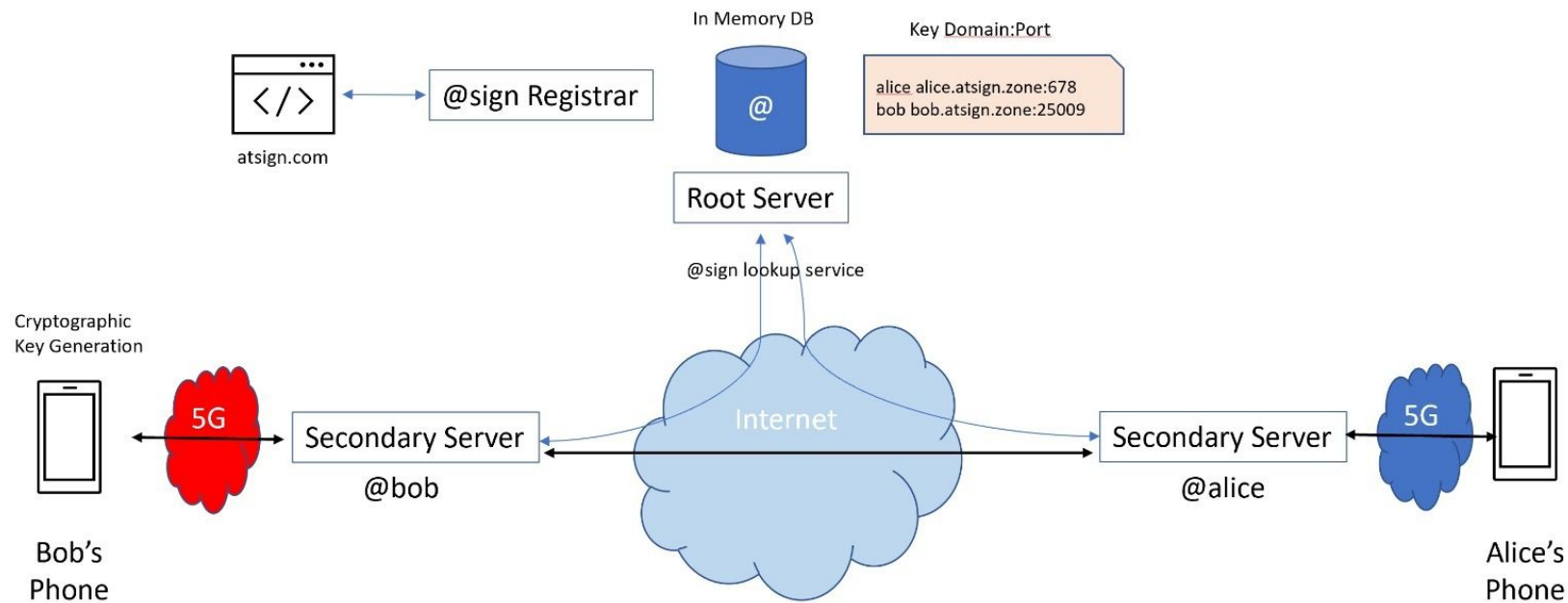
2020 AWS Instance Type Additions



49% of AWS EC2 instance additions in 2020 are
based on AWS Graviton2



We want to give people options on where their stuff is



JIT vs AOT

Dartshowplatform - A more useful 'Hello World!'

```
import 'dart:io' show Platform, stdout;  
  
void main() {  
    print(Platform.version);  
}
```

```
2.13.1 (stable) (Fri May 21 12:45:36  
2021 +0200) on "linux_arm64"
```



JIT - Just `dart run` it in the virtual machine

```
$ time dart run showplatform.dart
```

```
2.13.1 (stable) (Fri May 21 12:45:36  
2021 +0200) on "linux_arm64"
```

```
real    0m4.728s  
user    0m4.903s  
sys     0m0.684s
```



AOT - Compile it first then run the binary

```
$ dart compile exe showplatform.dart  
$ time ./showplatform.exe
```

```
2.13.1 (stable) (Fri May 21 12:45:36  
2021 +0200) on "linux_arm64"
```

```
real    0m0.023s  
user    0m0.008s  
sys     0m0.015s
```



Trade off - compilation is slow

```
$ time dart compile exe showplatform.dart
```

```
Info: Compiling with sound null safety  
Generated: showplatform.exe
```

```
real    0m17.434s  
user    0m20.912s  
sys     0m2.958s
```



Dependencies for building
and running an AOT binary

Dockerfile for dartshowplatform

```
FROM dart AS build
WORKDIR /app
COPY ./showplatform.dart .
RUN dart compile exe showplatform.dart \
    -o dartshowplatform

FROM scratch
COPY --from=build /runtime/ /
COPY --from=build /app/dartshowplatform \
    /app/dartshowplatform
ENTRYPOINT ["/app/dartshowplatform"]
```



Inside the Dockerfile for dart - what's in /runtime

```
RUN set -eux; \  
  for f in \  
    /etc/nsswitch.conf \  
    /etc/ssl/certs \  
    /lib/x86_64-linux-gnu/libc.so.6 \  
    /lib/x86_64-linux-gnu/libdl.so.2 \  
    /lib/x86_64-linux-gnu/libm.so.6 \  
    /lib/x86_64-linux-gnu/libnss_dns.so.2 \  
    /lib/x86_64-linux-gnu/libpthread.so.0 \  
    /lib/x86_64-linux-gnu/libresolv.so.2 \  
    /lib/x86_64-linux-gnu/librt.so.1 \  
    /lib64/ld-linux-x86-64.so.2 \  
    /usr/share/ca-certificates \  
  ; do \  
    dir="$(dirname "$f")"; \  
    mkdir -p "/runtime$dir"; \  
    cp --archive --link --dereference --no-target-directory "$f" "/runtime$f"; \  
  done
```



All a bit x86 - how do I get Arm?

atsigncompany/buildimage Dockerfile - arch specific

```
case "$(uname -m)" in \
  armv7l | armv7) ARCH="arm-linux-gnueabihf" ; \
    mkdir -p /runtime/lib/"$ARCH" ; \
    cp /lib/ld-linux-armhf.so.3 /runtime/lib/ld-linux-armhf.so.3 ; \
    cp /lib/arm-linux-gnueabihf/ld-linux-armhf.so.3 \
      /runtime/lib/arm-linux-gnueabihf/ld-linux-armhf.so.3 ;; \
  aarch64) ARCH="aarch64-linux-gnu" ; \
    mkdir -p /runtime/lib/"$ARCH" ; \
    cp /lib/ld-linux-aarch64.so.1 /runtime/lib/ld-linux-aarch64.so.1 ; \
    cp /lib/aarch64-linux-gnu/ld-linux-aarch64.so.1 \
      /runtime/lib/aarch64-linux-gnu/ld-linux-aarch64.so.1 ;; \
  *) ARCH="x86_64-linux-gnu" ; \
    mkdir -p /runtime/lib64/ ; \
    cp /lib64/ld-linux-x86-64.so.2 /runtime/lib64/ld-linux-x86-64.so.2 ;; \
esac && \
```



atsigncompany/buildimage Dockerfile - arch generic

```
for f in \  
    /etc/nsswitch.conf \  
    /etc/ssl/certs \  
    /lib/"$ARCH"/libc.so.6 \  
    /lib/"$ARCH"/libdl.so.2 \  
    /lib/"$ARCH"/libm.so.6 \  
    /lib/"$ARCH"/libnss_dns.so.2 \  
    /lib/"$ARCH"/libpthread.so.0 \  
    /lib/"$ARCH"/libresolv.so.2 \  
    /lib/"$ARCH"/librt.so.1 \  
    /usr/share/ca-certificates \  
; do \  
    dir="$(dirname "$f")"; \  
    mkdir -p "/runtime$dir"; \  
    cp --archive --link --dereference --no-target-directory "$f" "/runtime$f"; \  
done; \  

```



Cross platform building made easy

Dockerfile for dartshowplatform

```
FROM atsigncompany/buildimage AS build
WORKDIR /app
COPY ./showplatform.dart .
RUN dart compile exe showplatform.dart \
    -o dartshowplatform

FROM scratch
COPY --from=build /runtime/ /
COPY --from=build /app/dartshowplatform \
    /app/dartshowplatform
ENTRYPOINT ["/app/dartshowplatform"]
```



GitHub Action snippet

```
name: Build and push
id: docker_build
uses: docker/build-push-action@v2.5.0
with:
  file: ./dartshowplatform/Dockerfile
  push: true
  tags: |
    atsigncompany/dartshowplatform:automated
    atsigncompany/dartshowplatform:latest
    atsigncompany/dartshowplatform:GHA${{ github.run_number }}
  platforms: |
    linux/amd64
    linux/arm64/v8
```



The end results

TAG

automated

docker pull atsigncompany/dartshowpl... 

Last pushed 6 days ago by [atcpswan](#)

DIGEST	OS/ARCH	SCANNED	LAST PULL	COMPRESSE...
f37a5c8b28a8	linux/amd64	Disabled	a few seco...	4.51 MB
a2a5ea20f382	linux/arm64	Disabled	a few seco...	3.62 MB

https://github.com/atsign-company/at_dockerfiles

TAG

prod

docker pull atsigncompany/secondary:... 

Last pushed a day ago by [atcpswan](#)

DIGEST	OS/ARCH	SCANNED	LAST PULL	COMPRESSE...
460ddef70044	linux/amd64	Disabled	an hour ago	5.94 MB
35acf2f14200	linux/arm64	Disabled	an hour ago	5.04 MB



Recap

- Arm is important
 - ◆ 'Full Stack Dart' to complement Flutter Apps
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Questions?

Contacts

Chris Swan

chris@atsign.com

@cpswan

Anthony Prakash

Developer Relations

anthony@atsign.com

@anthony



The @ Company

At The @ Company (The At Company) we are technologists, creators, and builders with one thing in common: We love the Internet. You could go so far as to call us Internet optimists. Though we acknowledge that the Internet has deep flaws, we believe that we can extract all its goodness without sacrificing our privacy, time, and control over our digital identities.

We've committed ourselves to the creation of a more human Internet where privacy is a fundamental right and everyone owns their own data. Let's say goodbye to the fear and paranoia caused by data breaches and unsolicited online surveillance.

We've developed a fully-secure, privacy-compliant technology powering customer mobile applications: the @protocol, and with unique identifiers called @signs that act as keys into these experiences. With like-minded partners we're resolving these long-standing issues with a spirit of fun, not fear, and with delightful customer experiences.

FACT SHEET

Founded : 2019, by Barbara Tallent, Colin Constable, and Kevin Nickels, 30+ years executive experience at NCD/Netmanage, Credit Suisse, Deutsche Bank, Juniper Networks, Founded 3 prior startups, 3 exits. **Chairman**: Kim Perdikou, PCNET, Reader's Digest, most recently CIO, GM, EVP Juniper Networks

HQ : Virtual offices, with base in San Jose, CA.

Funding : \$11M seed, May, 2021

Employees: 22

Patents : 15 pending

Website : <https://atsign.com>